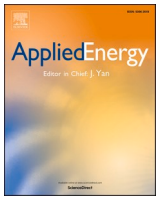
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Condition-triggered adaptive robust MPC for flexible load regulation in energy-intensive industries: A case study of aluminum electrolysis

Jinbo Nie [a,b], Qiaojie Bai [a,b], Congwei Tang [a,b,c], Jinlong Du [a,b,*], Jianhang Hu [a,b,c,d,*], Wei Cheng [e], Cheng Tan [a,b], Ruijin Fan [a,b], Yong Yu [a,b], Wanzhang Yang [e], Hua Wang [a,b,c]

[a] Engineering Research Center of Metallurgical Energy Conservation and Emission Reduction, Ministry of Education, Kunming University of Science and Technology, Kunming 650093, PR China
[b] School of Metallurgical and Energy Engineering, Kunming University of Science and Technology, Kunming 650093, PR China
[c] Southwest United Graduate School, Kunming 650092, PR China
[d] State Key Laboratory of Advanced Metallurgy for Non-Ferrous Metals, Kunming Metallurgy Research Institute, Kunming 650503, China
[e] Yunnan Aluminum Co., LTD, Kunming 650502, PR China

HIGHLIGHTS

- Proposes a condition-triggered robust MPC framework for energy-intensive industries.
- Modular Energy Hub architecture enables transferability to steel and cement sectors.
- Conformal prediction calibrates uncertainty bounds without distributional assumptions.
- A 300-kA aluminum electrolysis line validates feasibility and cost cuts.

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ABSTRACT

Energy-intensive industries face a growing conflict between power-system flexibility requirements and the stringent stability constraints of continuous electro-thermal processes. To overcome the limited adaptability of static scheduling and controllers with fixed conservatism under multi-regime operating conditions, this study proposes a condition-triggered robust model predictive control (CT-RMPC) framework. The method couples a multi-energy hub representation with distribution-free conformal prediction to calibrate uncertainty bounds and introduces a trigger layer that adjusts the robust budget Γ online according to regime indicators, thereby trading off economic performance and risk-aware feasibility. Industrial validation is conducted on a 300kA aluminum electrolysis line using 18 months of high-resolution operational data. The proposed controller reduces the peak grid import power from 160 MW to 151 MW (−5.6%) while maintaining the specific energy consumption within 13.2 ± 0.2 MWh•t^{-1} Al. The validated period yielded a cumulative cost savings of 652,000 USD, and the monthly unit net effect remains positive, ranging from 0.4 to 3.8 USD·t^{-1}Al, with demand-charge reduction contributing more than 35% of the total benefit. Although validated on aluminum electrolysis, the framework is formulated in a modular manner and can be extended to other EIIs with strong thermal inertia and multi-energy coupling, provided that process-specific states and feasibility constraints are re-instantiated and validated.

1. Introduction

The integration of renewable energy is increasingly transforming decarbonization from a supply-side problem into a system-wide operational challenge, where demand-side flexibility becomes essential for maintaining reliability and cost-effectiveness of power systems [1,2]. Recent reviews further emphasize that demand-side flexibility is indispensable for integrating high shares of variable renewables and maintaining system reliability. In this context, energy-intensive industries

* Corresponding authors at: Engineering Research Center of Metallurgical Energy Conservation and Emission Reduction, Ministry of Education, Kunming University of Science and Technology, Kunming 650093, PR China.
E-mail addresses: jinlongdu@kust.edu.cn (J. Du), hujh51@126.com (J. Hu).

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Nomenclature

Abbreviations

CI	Carbon Intensity
COP	Coefficient of Performance
CT-RMPC	Condition-Triggered Robust Model Predictive Control
EH	Energy Hub
EIIs	Energy-Intensive Industries
HP	Heat Pump
LSTM	Long Short-Term Memory
MPC	Model Predictive Control
MPIW	Mean Prediction Interval Width
NE	Net Effect
PICP	Prediction Interval Coverage Probability
PV	Photovoltaic
SCP	Split Conformal Prediction
SEC	Specific Energy Consumption
SoC	State of Charge of TES
TES	Thermal Energy Storage

Symbols

t	Index for time periods, $t \in \{1, \ldots, H\}$
s	Index for piecewise linear segments of heat pump COP
i	Index for data samples or scenarios
$\mathscr{U}$	Uncertainty set for robust optimization
Γ	Uncertainty budget parameter
λ_d	Demand billing coefficient, USD·kW^{-1}·month^{-1}
$\delta_{s,t}$	Segment selection variable for COP at time t
C_k	The k-th operating-condition class
α_k	Post-calibration error level associated with class C_k
$\Gamma_{Q,k}$	Γ-budget robustness parameter for load uncertainty under class C_k
$\Delta I_{max,k}$	Upper bound of DC current ramping under class C_k
R_k^{cell}	Cell operational flexibility/ramp-related parameter under class C_k
P_t^{PV}	Photovoltaic power generation, MW
P_t^{buy}	Power purchased from grid, MW
P_t^{sell}	Power sold to grid, MW
P_t^{HP}	Electrical power consumption of Heat Pump, MW
$P_{AC,t}^{cell}$	AC power consumption of electrolytic cells, MW
P_t^{aux}	Auxiliary power load, MW
$P_{HP}(t)$	Heat pump electrical input power at time t, MW
$P_{cell}(t)$	DC power delivered to the electrolytic cell at time t, MW
$P_{rect,in}(t)$	Input AC power to the rectifier at time t, MW
P_t^{pv}	PV generation power at time t, MW
$\underline{P_t}$	Lower bound of conservative interval forecast forP_t^{pv}, MW
$\overline{P_t}$	Upper bound of conservative interval forecast for P_t^{pv}, MW
Q_t^{HP}	Thermal power supplied by the heat pump at time t, MW
Q_t^{dis}	Thermal discharge power of the TES at time t, MW
Q_t^{ch}	Thermal charging power of TES at time t, MW
Q_t^{dump}	Dumped/curtailed thermal power at time t, MW
Q_t^{dem}	Total heat demand to be satisfied at time t, MW
$Q_{HP}(t)$	Heat pump thermal output power at time t MW
Q_t^{load}	Heating/thermal load demand at time t, MW
$\underline{Q_t}$	Lower bound of conservative interval forecast for Q_t^{load}, MW
$\overline{Q_t}$	Upper bound of conservative interval forecast for Q_t^{load}, MW
u_t^{ch}	Charging state indicator of TES at time t, MW
u_t^{dis}	Discharging state indicator of TES at time t, MW
$\mathscr{U}_Q(\Gamma)$	Γ-budget uncertainty set for the heating load
η_{ch}	Charging efficiency of TES
η_{dis}	Discharging efficiency of TES
η_{rect}	Rectification efficiency
η_t	Load deviation factor within the interval

(EIIs)—including steel, cement, and chemicals—constitute a dominant share of electricity and fuel consumption, and their ongoing electrification further amplifies grid-coupled load magnitude and variability [3,4]. Meanwhile, many EII production lines operate as tightly coupled continuous processes with narrow safety margins, where aggressive load modulation may induce electro-thermal instability, quality deviations, or even safety risks [5]. Consequently, EIIs face a critical dilemma of reconciling increasing grid demand for load flexibility with the strict stability requirements of complex industrial processes, which calls for a scalable control paradigm capable of coordinating flexibility provision with guaranteed process feasibility [6].

Deep demand response pushes industrial processes beyond the nominal region for which most control loops are tuned, and the resulting regime switching (e.g., rapid ramping, load shedding, or price-driven rescheduling) magnifies model–plant mismatch. Conventional PID and linear MPC remain effective near the design point [7]. However, large excursions (e.g., rapid ramping, load shedding, or price-driven rescheduling) may trigger regime transitions and amplify model–plant mismatch, making static optimization and fixed-parameter control brittle in practice. Accordingly, many industrial energy management implementations still rely on rule-based dispatch or deterministic real-time optimization, which often cannot provide a systematic economy–risk balance under time-varying uncertainty [8]. Stochastic MPC has been introduced to enhance resilience against disturbances via chance-constraint-based formulations, yet its performance depends on the fidelity of uncertainty descriptions [9]. Robust-optimization-based MPC avoids explicit distributional assumptions, but the design and calibration of uncertainty sets or budgets directly govern conservatism and tractability, which becomes particularly challenging under nonstationary conditions [10,11]. To obtain reliability-calibrated uncertainty bounds without specifying a data distribution, conformal prediction has recently been explored to construct empirically valid prediction intervals for renewable uncertainty in operational decision-making [12]. Meanwhile, event-triggered robust MPC highlights the potential of condition-dependent updating and adaptation mechanisms in closed-loop control, motivating adaptive robustness that reacts to operating conditions rather than remaining static [10]. Recent advancements have demonstrated the substantial flexibility potential across diverse energy-intensive sectors. For instance, Li et al. integrated process-level scheduling into bidding strategies for steelmaking to coordinate market trading with production [13]. Similarly, a robust optimization framework was developed for cement plants to mitigate the risks of market volatility, while the real-time demand response capabilities and carbon performance of hot rolling processes have been quantitatively assessed [14,15]. Notably, recent contributions in related journals have emphasized flexibility exploitation under high renewable penetration and carbon constraints. However, most existing studies focus primarily on scheduling-level coordination or static robust optimization, with limited attention to regime-aware adaptive control capable of addressing frequent operating condition transitions and nonlinear process dynamics. These limitations highlight the need for a condition-triggered adaptive robust MPC framework that systematically integrates uncertainty calibration with multi-regime operational adaptation in energy-intensive processes.

Against this backdrop, this work proposes a generalized CT-RMPC framework. By coupling a multi-energy hub representation with

conformal-prediction-calibrated uncertainty bounds [1], the proposed controller introduces a dynamic trigger layer that detects regime shifts and updates the robust budget Γ online, enabling the system to switch between economy-prioritized and safety-conservative modes while maintaining constraint satisfaction [12]. The framework is validated on an industrial aluminum electrolysis line, a representative electricity-dominant process characterized by strong electro-thermal coupling and stringent safety constraints, which makes it a challenging yet relevant testbed for assessing robustness and feasibility under flexible operation [16,17]. Aluminum electrolysis is selected as the testbed because it represents a highly constrained scenario due to its pronounced electro-thermal coupling and narrow safety margins. If the framework works here, it is applicable to less constrained EII processes.

While aluminum electrolysis serves as the validation testbed, the framework targets EIIs more broadly, where controllable but safety-critical loads are increasingly expected to provide grid services. The key ingredients do not depend on aluminum-specific chemistry. They apply equally to other EIIs with comparable characteristics, such as steel reheating furnaces, cement kilns, chlor-alkali electrolysis, or large-scale industrial refrigeration Table 1 summarizes the generic modules and the process-specific interfaces for such deployment. To position CT-RMPC within existing research on industrial flexibility, the following paragraphs briefly review energy-hub-based multi-energy modeling and MPC-based uncertainty-aware scheduling and control.

To systematically model the internal flexibility potential of such strongly coupled industrial processes, the concept of multi-energy complementarity and Energy Hub (EH) is adopted as the underlying modeling architecture [18]. The EH theory, first proposed by Geidl et al., provides a unified matrix formulation to describe the conversion and storage of multiple energy carriers [19]. Subsequently, this concept was expanded to model an energy-consuming entity, such as a factory or industrial park, as an integrated system of multiple energy inputs, conversion devices, and storage units [20]. Within this framework, decoupling and spatiotemporal transfer among different energy carriers (e.g., electricity, heat, and gas) are facilitated to break the rigidity of single-energy loads [21]. For instance, using heat pumps (HP) together with thermal energy storage (TES) to achieve electricity–heat decoupling is a key technology for enhancing industrial flexibility [22].During low-price periods, HP-driven heat production and TES charging can be scheduled, whereas TES discharges during high-price periods to satisfy thermal demand and reduce grid electricity consumption [23,24]. Related studies on multi-energy distributed energy systems have shown that coordinating flexible measures such as HP and TES through price-driven demand response can reduce operating costs and improve grid load profiles [25].

To enable such transferability and systematic model, researchers have extensively studied optimal scheduling for industrial energy hubs. Prior research treated the problem as deterministic, assuming precise forecasts for load, electricity prices, and renewable outputs [8]. In reality, however, the output of PV and wind power is characterized by significant uncertainty and volatility [26]. Optimization based on inaccurate point predictions poses a high risk of causing the system to deviate from its planned operation, leading to economic losses and even safety risks [27]. Due to its closed loop, rolling optimization characteristics, model predictive control is widely employed in the real-time scheduling of EH to address uncertainty [28]. Combining uncertainty hand closed loops, MPC can be divided into stochastic MPC (SMP) and robust MPC (RMPC). SMP relies on precise descriptions of the probability distribution of uncertainties [29]. However, it is extremely difficult to obtain an accurate probability distribution. In contrast, Robust Optimization (RO) does not rely on probability distributions, but instead constructs an uncertainty set and seeks solutions that satisfy the constraints in all possible worst-case scenarios [30]. RMPC integrates Robust Optimization (RO) into the MPC framework to enhance resilience against prediction errors. RMPC has been increasingly explored in safety-critical industrial applications [31]. However, traditional RMPC approaches, which pursue absolute worst-case scenarios, tend to yield excessively conservative decisions, resulting in unnecessary economic losses. To balance robustness with economic efficiency, uncertainty-budgeted Γ-robust optimization frameworks have gained widespread adoption in recent years for multi-energy systems and industrial load optimization. Such budgeted robust methods employ an adjustable budget parameter Γ to control the degree of conservatism, enabling decisions to strike a balance between excessive caution and high risk [32,33]. In particular, Γ restricts the number of time steps within a horizon that can simultaneously reach their worst-case boundaries, while the remaining steps are assumed to stay at their nominal values. However, conventional Γ-robust MPC typically adopts a fixed budget, making it difficult to reconcile economic performance under normal conditions with risk-aware feasibility under abnormal regimes and internal operating variations [34].

Despite significant progress in the research, substantial gaps remain when applying multi-energy-coupled energy recovery and robust multi-stage control to safety-critical industrial processes characterized by strong non-linear coupling and strict operational windows. While primary aluminum electrolysis serves as a representative case due to its extreme complexity, the lack of generalized control strategies for such multi-regime processes remains a broad gap. Existing industrial energy recovery studies predominantly simplify industrial loads into arbitrarily dispatchable black boxes, entirely disregarding their intricate internal process constraints [21,24]. Similarly, existing studies targeting specific EII sub-sectors (e.g., steel, aluminum) often optimize in isolation; they typically focus solely on internal electro-chemical or thermo-physical balance, lacking a systematic view of the system as an EH to be synergistically optimized with multi-carrier flows such as HP, TES, and PV [35,36]. Furthermore, conventional Γ-robust MPC approaches typically rely on a fixed uncertainty budget, making it difficult to reconcile economic efficiency under normal conditions with rigorous safety under abnormal ones [37,38]. Therefore, a framework that adaptively adjusts Γ based on the risk level is essential to achieve both safety and economic optimization for the multi-energy system [39,40].

This study makes three main contributions. First, this work develops a transferable energy-hub formulation for electricity-dominant electro-thermal industrial processes, explicitly capturing multi-physics energy conversion, HP–TES coordination, and process-feasibility constraints, thereby linking multi-energy dispatch with safety-relevant operating envelopes. Secondly, an LSTM-based forecasting module is integrated with distribution-free conformal prediction to generate reliability-calibrated uncertainty intervals for external disturbances without assuming a specific probability distribution; a dual-based robust counterpart is further derived to ensure tractable rolling optimization under Γ-budgeted uncertainty. Thirdly, this work proposes a condition-triggered adaptive Γ adjustment mechanism that updates conservatism online according to real-time operating conditions and internal state changes (e.g., discrete process disturbances), enabling an explicit economy–risk trade-off that supports peak shaving and carbon-aware operation while maintaining process safety.

Table 1
Transferability of CT-RMPC: generic modules and process-specific interfaces.

Layer	Process-specific interfaces to be instantiated when transferring
Multi-energy modeling	Carrier and device set; efficiency and COP curves; capacity and availability limits; coupling topology
Process feasibility	Safety-relevant process states and safe bounds; internal event definitions
Uncertainty calibration	Prediction targets and features; nonconformity score; rolling calibration window
Robust optimization	Robust-protected constraints; (Γ_{min}, Γ_{max}); scaling settings
Trigger & adaptation	Indicator selection; thresholds and hysteresis; risk metric definition
KPIs & implementation	Safety KPI definition and reporting units; carbon boundary and grid emission factor scenarios

2. Methods

To reconcile the conflicting requirements of grid flexibility and process stability, this study establishes a systematic control framework. The work first formulates a generalized EH model applicable to a broad class of electro-thermal industrial processes. Subsequently, the work introduces a condition-triggered robust control strategy to manage multi-regime dynamics.

To operationalize transferability beyond aluminum electrolysis, CT-RMPC is formulated as a modular architecture that separates an algorithmic core from process instantiation. The EH provides a generic representation of multi-carrier energy balances and the couplings introduced by conversion units and storage devices. When transferring the framework to a new energy-intensive industrial process, one specifies the available energy carriers, conversion and storage units, coupling topology, and device performance characteristics, together with implementable operational limits and ramping constraints. Process safety is ensured by selecting a parsimonious set of safety-relevant state variables and proxy indicators and encoding the corresponding feasibility envelope as hard constraints, including any discrete internal events that may trigger regime changes. Key disturbances, including electricity prices, renewable generation, and demand, are forecast and calibrated using distribution-free conformal prediction to obtain target-coverage uncertainty intervals, which are embedded into the Γ-budget robust MPC formulation. A condition-trigger mechanism then updates Γ online according to risk indicators, enabling an explicit economy–conservatism trade-off while maintaining feasibility. Carbon-related outcomes are evaluated only as an optional scenario extension when time-varying grid emission factors or carbon prices are available.

2.1. System configuration and case study description

In this study, flexible load regulation is defined as the controllable adjustment of the net grid power exchange within the EH by coordinating auxiliary load shifting, modulating HP electrical input, and managing the charging/discharging of TES. Importantly, the primary electrolytic cells are operated within their prescribed stable power window; therefore, flexibility is mainly harvested from the auxiliary electro-thermal subsystems and is only reflected to the grid as a reshaped net demand profile. Such flexibility is strictly constrained by the allowable ramping limits and the electro-thermal stability requirements of the electrolysis process, which jointly bound the achievable regulation depth and duration.

To validate the proposed generalized control framework for EIIs, this study selects the aluminum electrolysis process as a representative testbed. This process is chosen because it exemplifies a highly constrained electro-thermal scenario characterized by high energy demand and strictly bound safety margins. Utilizing 18 months of high-frequency measurement data from a large-scale aluminum smelter, this study implements a Γ-robust MPC framework capable of adapting to varying operating conditions and coordinating multi-energy systems involved in the production process.

Fig. 1 illustrates the specific EH structure of the case study plant. The system integrates multiple energy flows, including on-site photovoltaic generation, grid electricity, natural gas, and supporting services. The diagram highlights the electro-thermal coupling mechanism. Specifically, the HP serves as a conversion bridge. It recovers low-grade waste heat from the electrolysis exhaust to charge the TES. Finally, Fig. 1 establishes the closed-loop logic of the proposed control strategy. In contrast to traditional open-loop scheduling, the trigger-mode feedback loop indicated by the grey dashed line monitors real-time process variations and feeds this status back to the Γ -robust MPC controller. This

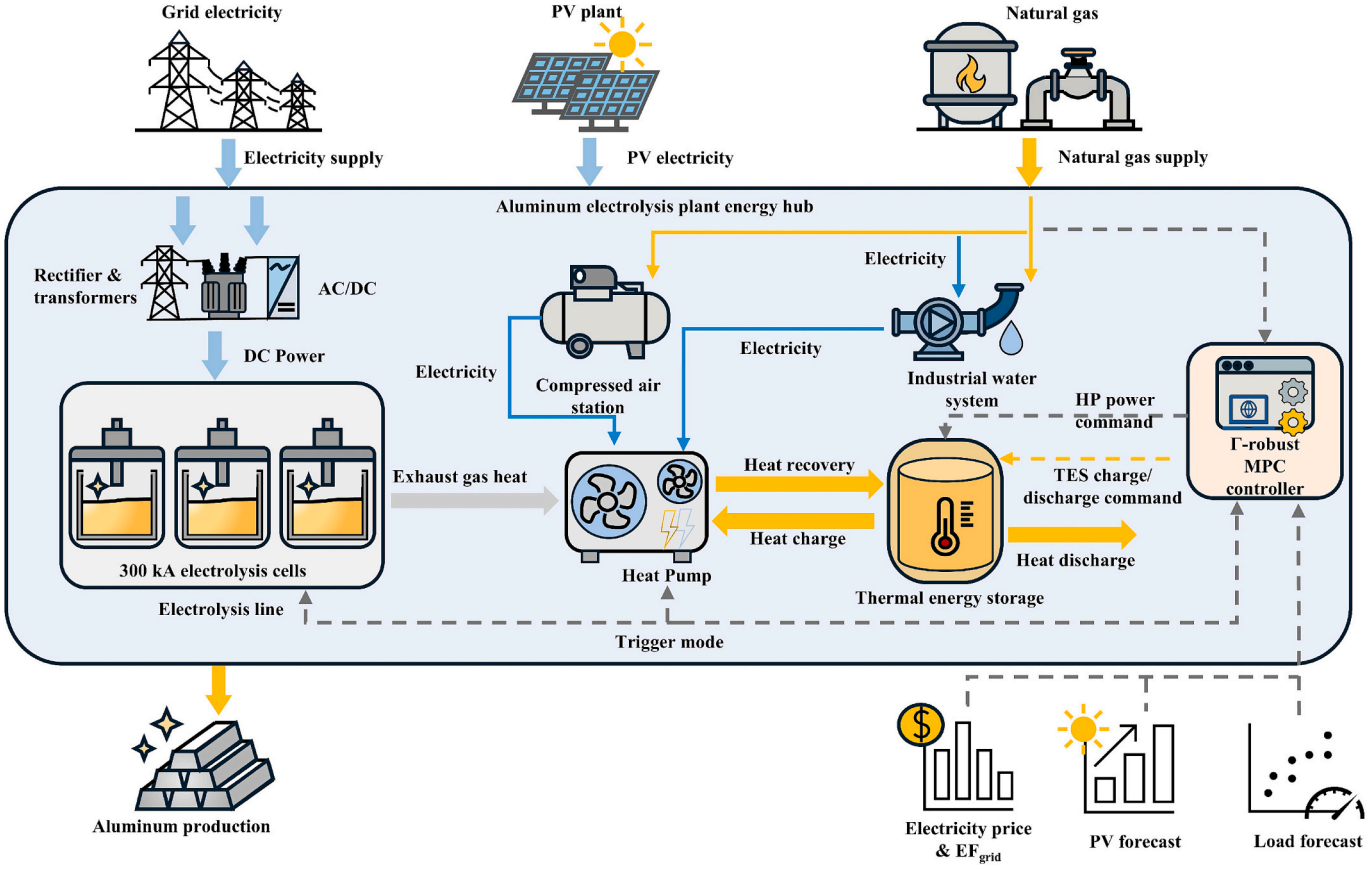


Fig. 1. System boundary and energy hub structure of the case study plant.

feedback mechanism enables the controller to adaptively switch between economic optimization and safety-prioritized modes, thereby ensuring that flexible dispatch does not compromise the stability of the core electrolysis process.

The empirical validation in this research relies on 18 months of operational data acquired from a major aluminum smelter in Yunnan Province, which operates over one hundred 300 kA reduction cells. First, this work collected 18 months of 5-min-resolution operational records for power and major multi-energy flows. This high-resolution dataset is suitable for intraday MPC simulations, whilst data for the remaining periods is aggregated into monthly summaries to support cross-month NE statistical analysis and robustness validation. Furthermore, the electricity pricing data are sourced from China Southern Power Grid. These tariffs adhere to real-time and time-of-use pricing curves. Data processing entails resampling these external time series onto a unified five-minute time axis and synchronizing them with plant-level operational data, where missing values are interpolated according to established rules and night-time photovoltaic generation is set to zero. Finally, the dataset is divided chronologically into training, validation, calibration, and test sets. The calibration set is used solely for adjusting the conformity interval parameters, with the specific methodology outlined in the following section.

2.2. Disturbance forecasting and interval construction

To address the stochasticity inherent in industrial energy hubs—stemming from both renewable generation and internal process load fluctuations—a data-driven uncertainty quantification scheme is employed. The LSTM network is utilized for point forecasting of key disturbances. However, given that point forecasts fail to effectively capture random errors, the Split Conformal Prediction (SCP) framework is adopted to quantify uncertainty. As a nonparametric approach, SCP constructs prediction intervals with rigorous coverage guarantees. Crucially, it utilizes residuals derived from a dedicated calibration set that is strictly independent of both the training and testing sets, thereby quantifying uncertainty without assuming any specific error distribution. In this study, the nominal coverage level is 90%, with a high-reliability mode set at 95%. Using a preset confidence level, SCP produces reliable prediction intervals directly from past prediction errors [41]. The quality of these intervals is evaluated by using two standard metrics.

Prediction interval coverage (PICP) measures the proportion of observed values that fall within the interval, with the objective of matching the preset confidence level. Mean prediction interval width (MPIW) quantifies interval compactness. For a given target coverage rate, narrower intervals correspond to reduced conservatism, thereby enhancing scheduling utility. Thus, the calibrated interval sequence serves as the uncertainty bound for the robust MPC.

Conformal prediction offers two main advantages over traditional parameter-based confidence intervals. First, it guarantees coverage with limited samples without assuming a specific data distribution, rendering it particularly effective for small-sample settings [42]. Secondly, it is model-agnostic and can be combined directly with deep-learning predictors such as LSTM [43]. Recent work in photovoltaic power forecasting demonstrates that conformal prediction intervals outperform Gaussian-assumption intervals in both coverage and width [42]. These properties underpin the interval-robust MPC proposed in this work, supporting its feasibility under variable operating conditions.

2.3. Energy hub modeling and constraints

While the proposed framework is applicable to generic electro-thermal loads, explicit mathematical modeling is necessary for implementation. Here, this work instantiates the general EH constraints specifically for the aluminum electrolysis process to capture its unique rectification and thermal characteristics. Reliability-aware constraints are explicitly embedded, covering electricity-heat coupling and equipment limitations. Thus, the system satisfies hard limits as follows:

The electrical side of the EH must always maintain supply–demand balance. The relationship describing the power balance among photovoltaic output, grid power exchange, and equipment load is expressed as follows:

$$P_t^{PV} + P_t^{buy} - P_t^{sell} = P_t^{HP} + P_{AC,t}^{cell} + P_t^{aux} \tag{1}$$

where P_t^{PV} denotes the predicted photovoltaic generation (MW), P_t^{buy} and P_t^{sell} represent the power purchased from and sold to the grid (MW), respectively, P_t^{HP} is the electrical power consumption of the HP (MW), $P_{AC,t}^{cell}$ is the AC-side power consumption of the electrolytic cells (MW), and P_t^{aux} denotes other auxiliary power loads (MW).

The thermal side must also satisfy the supply–demand balance. The system thermal output, comprising the heating power of the HP and the heat released by the TES, must equal the combined heating demand of the electrolysis process and the factory auxiliary systems:

$$Q_t^{HP} + Q_t^{dis} - Q_t^{ch} - Q_t^{dump} = Q_t^{dem}, Q_t^{dump} \geq 0 \tag{2}$$

where Q_t^{HP} is the thermal power supplied by the HP (MW); Q_t^{dis} and Q_t^{ch} denote the thermal discharge and charging power of the TES (MW), Q_t^{dump} represents the dissipated or curtailed thermal power (MW), and Q_t^{dem} denotes the total thermal load demand to be satisfied in that period (MW). Eq. (2) states that the combined output of the HP and the TES must meet the heat demand of the user. Surplus generation is stored in the TES, whereas deficits are supplied by heat discharged from the TES.

The thermal energy stored and released by the TES follow thermodynamic laws. The TES state evolves at each time step driven by charging and discharging dynamics. The governing equation is expressed as follows:

$$E_{t+1} = E_t + \eta_{ch} Q_t^{ch} \Delta t - \frac{1}{\eta_{dis}} Q_t^{dis} \Delta t \tag{3}$$

where η_{ch} and η_{dis} denote the heating and heat-release efficiencies, respectively. Eq. (3) describes the energy state of the TES between t and t + 1. Eq. (4) constrains the *SoC* within the specified upper and lower bounds to prevent overcharging or overdischarging.

$$E_{min} \leq E_t \leq E_{max}, E_t \geq \rho_{min} E_{max} \tag{4}$$

where E_t denotes the energy state of the TES at time t (MWh), η_{ch} and η_{dis} represent the heating and heat-release efficiencies, respectively, and Δt is the duration of the scheduling time step (h).

HP efficiency typically depends on ambient temperature. Here, the HP coefficient of performance (COP) is represented via a piecewise linear approximation. In particular, the relationship between output thermal power $Q_{HP}(t)$ and input electrical power $P_{HP}(t)$ is:

$$Q_{HP}(t) = \sum_s COP_s \cdot \delta_{s,t} \cdot P_{HP}(t), \delta_{s,t} \in \{0, 1\} \tag{5}$$

where COP_s is the coefficient of performance for the s-th linear segment, and $u_{s,t}$ is a binary decision variable indicating whether the s-th COP segment is selected at time t.This model represents the nonlinear variation of HP efficiency through a linear combination approach, adapting to performance adjustments under different operating conditions.

The rectifier converts alternating current into direct current to supply the electrolytic cell. The rectification efficiency is η_{rect}. Therefore, the relationship between the direct current power obtained by the electrolytic cell and the input alternating current power is:

$$P_{cell}(t) = \eta_{rect} \cdot P_{rect,in}(t) \tag{6}$$

where $P_{cell}(t)$ denotes the DC power obtained by the electrolytic cell (MW), η_{rect} is the rectification efficiency, and $P_{rect,in}(t)$ represents the

input AC power to the rectifier at time t (MW).

The power output of the electrolytic cell also needs to meet the bandwidth and ramping constraints. The output power of the electrolytic cell is limited by its rated power range and can be expressed as:

$$P_{\text{cell}}^{min} \le P_{\text{cell}}(t) \le P_{\text{cell}}^{max} \tag{7}$$

where P_{cell}^{min} and P_{cell}^{max} denote the rated lower and upper AC power consumption limits of the electrolytic cell (MW). In particular, the power regulation of the electrolytic cell must comply with strict ramping constraints. Excessive power fluctuations disrupt the electro-thermal balance within the cell. This disruption causes sharp changes in electrolyte temperature. In extreme cases, this instability triggers anode effects or causes physical damage to the cell.

The charging and discharging process of the TES system needs to satisfy the mutual exclusion constraint. To be precise, the TES cannot simultaneously charge and discharge. For this purpose, binary decision variables u_{ch} and u_{dis} are introduced to represent the charging and discharging states, and the mutual exclusion constraint is:

$$\begin{aligned} 0 \le Q_t^{ch} &\le u_t^{ch}\overline{Q}^{ch} \\ 0 \le Q_t^{dis} &\le u_t^{dis}\overline{Q}^{dis} \\ u_t^{ch} + u_t^{dis} &\le 1 \\ u_t^{ch}, u_t^{dis} &\in \{0,1\} \end{aligned} \tag{8}$$

where u_t^{ch} and u_t^{dis} are binary decision variables (0 or 1) representing the charging and discharging state indicators of the TES; while $\overline{Q}^{ch}$ and $\overline{Q}^{dis}$ denote the maximum charging and discharging power capacity limits (MW).

Ensuring process safety forms the primary premise for all constraints in this work. This fundamental requirement drives the selection of robust MPC instead of deterministic MPC.

2.4. Robust MPC optimization strategy

After defining the constraint layer, the problem is formulated as a worst-case optimization problem with budget constraints. The Γ -budget robust set is employed to enhance solvability, and dual reformulation is applied to the worst-case scenario. Ultimately, this formulation establishes a roll-out executable MPC model.

As shown in Fig. 2, after obtaining the interval prediction results described in Section 2.2, the perturbation intervals were embedded into the Γ -budget robust MPC framework, and a solvable rolling optimization model is implemented through Dual reformulation. External disturbances such as heating load Q_t^{load}, electricity price π_t, and photovoltaic P_t^{pv} were regarded as interval uncertain parameters. At each step time t, the system is subject to the following constraints:

$$\begin{aligned} Q_t^{\text{load}} &\in \left[\underline{Q_t}, \overline{Q_t}\right] \\ P_t^{\text{pv}} &\in \left[\underline{P_t}, \overline{P_t}\right] \end{aligned} \tag{9}$$

the underline and double overline symbols denote the lower and upper bounds generated by the conservative forecast, respectively. The model leverages this interval uncertainty to construct an interval-robust MPC framework. The framework imposes control constraints on each disturbance scenario within the prediction interval. By optimizing for these worst-case disturbances, the MPC framework enhances the reliability of the operational strategy [44].

The proposed robust MPC model adopts the Γ -budget robustness mechanism. This approach specifically addresses the uncertainty of the heating load. Within the rolling time domain, the worst-case value appears for a maximum of Γ moments. A larger Γ results in a more conservative strategy. Based on this principle, the formula defines the uncertainty set:

$$\mathscr{U}_Q(\Gamma) = \left\{ Q_{load,t} = \underline{Q_t} + \eta_t\left(\overline{Q_t} - \underline{Q_t}\right), 0 \le \eta_t \le 1, \sum_{t=1}^{H} \eta_t \le \Gamma \right\} \tag{10}$$

where H represents the total number of time periods for MPC rolling optimization, and $\eta_t \in [0,1]$ is an auxiliary variable indicating the degree of load deviation. $\eta_t = 0$ corresponds to the predicted lower bound $\underline{Q_t}$, while $\eta_t = 1$ corresponds to the upper bound $\overline{Q_t}$. Constraint $\sum_{t=1}^{H} \eta_t \le \Gamma$ restricts the cumulative number of times the upper bound load occurs simultaneously in any considered situation to no more than Γ time periods. Through the setting of this uncertain set, robust optimization ensures security while avoiding excessive conservatism caused by taking the worst-case scenario. A similar Γ -budget framework can also be applied to other uncertain parameters. For instance, the uncertainty of real-time electricity prices can be robustly modeled by limiting the

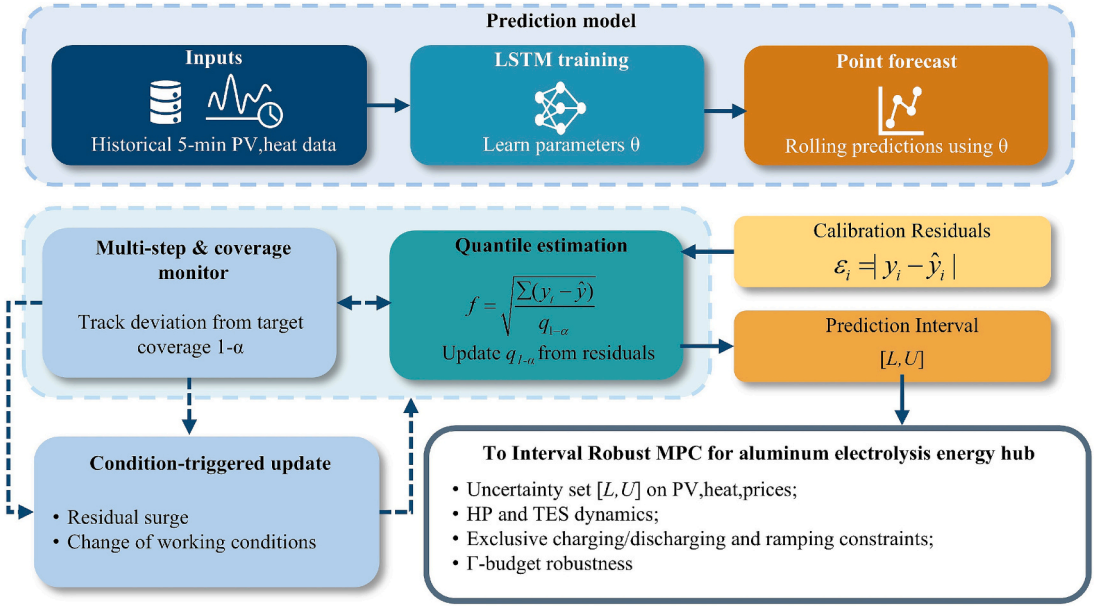


Fig. 2. Flowchart of the integrated process comprising prediction, calibration, and condition triggering.

number of times they experience the most extreme fluctuations [45].

Based on the uncertainty set Eq. (10), the model formulates a min–max robust optimization problem, which is computationally intractable in its raw form. This issue is addressed by applying strong duality theory to the inner worst-case maximization subproblem. The formulation introduces non-negative dual variables. These variables correspond to the uncertainty budget and bound constraints. Consequently, this process replaces the inner maximization operator. It effectively converts the problem into a set of deterministic linear constraints. This transformation effectively converts the original bi-level problem into a standard, single-level linear program or mixed-integer linear program (MILP) that can be solved efficiently by off-the-shelf solvers without iterative worst-case searching. Because a single static Γ cannot cover both normal and abnormal operating conditions, the next section introduces an operating-condition trigger layer. This layer dynamically adapts Γ and coverage levels based on assessed risks, utilizing parameter sets derived from historical data analysis to balance safety and economy.

Specifically, the inner worst-case maximization is linear in the uncertain parameters and is posed over a Γ-budget uncertainty set, under which the dual reformulation is valid under standard feasibility and boundedness conditions. Strong duality can therefore be invoked to replace the inner maximization with its dual, yielding an equivalent single-level deterministic program with zero duality gaps. When the scheduling layer contains only continuous decision variables, the resulting formulation is LP. When binary variables are introduced, the same reformulation leads to MILP. The computational complexity and solution efficiency of the CT-RMPC framework are critical for its real-time deployment under a 5-min dispatch cycle. The reformulated optimization problem is expressed as a single-level MILP, which can be efficiently handled by mature branch-and-bound algorithms implemented in commercial solvers such as Gurobi. Computation time was evaluated on a standard industrial workstation, and the average solution time per control step is approximately 2.4 s, with the most complex regime-switching cases requiring no more than 9.5 s. These results remain well below the 5-min operational interval, confirming the practical feasibility of the proposed approach. Computational efficiency primarily benefits from the structured condition-triggered mechanism, which restricts uncertainty adaptation to a finite set of predefined regimes rather than performing iterative worst-case searches. Consequently, the framework maintains stable convergence behavior and sufficient temporal margins for real-time industrial implementation.

2.5. Condition-triggered mechanism and event classification

A static uncertainty budget Γ inevitably faces a fundamental trade-off: a high value ensures worst-case safety at the expense of economic performance, while a low value prioritizes cost savings, but risks constrain violations under disturbance. To resolve this conflict, this work proposes a hierarchical trigger layer that dynamically adapts the control mode based on the real-time system operational state. This mechanism allows the controller to shift between yield-maximizing and risk-averse behaviors autonomously. Although specific triggers vary across different industrial sectors, they can be theoretically unified into three generic categories. The first category comprises external market events, which are driven by exogenous signals such as grid price fluctuations or dispatch commands. The second category includes discrete internal events, referring to deterministic maintenance operations, such as equipment switching or ramping, that temporarily alter the system topology. The third category covers stochastic process anomalies, which capture random, non-deterministic deviations in state variables indicating potential instability or subsystem faults.

To improve the transferability of the trigger mechanism beyond the aluminum electrolysis case, the condition-triggered layer is formulated through a systematic procedure that integrates expert knowledge with data-driven analysis. For a new industrial process, candidate trigger indicators are first selected from measurable market signals, internal operational states, and key process stability variables. Historical high-resolution data are then used to statistically evaluate their relevance to risk-related outcomes such as constraint violations or abnormal state deviations. Trigger thresholds are quantified using reproducible rules (e.g., quantile-based criteria or validation-based tuning), and hysteresis logic can be introduced to prevent frequent switching. Finally, the identified trigger conditions are periodically recalibrated using rolling data windows to account for nonstationary operating environments while preserving safety margins. In the context of our aluminum electrolysis case study, to verify the effectiveness of the proposed framework, the case studies consider seven representative operating conditions $\mathscr{C} = \{C_0, \ldots, C_6\}$, categorized into market regimes, internal operations, and process anomalies, as detailed in Table 2. Upon detection of a specific condition, the uncertainty budget Γ and associated safety margins are automatically updated to the pre-calibrated levels defined in Eq. (11). This adaptive approach ensures that the system maintains a context-aware balance between robustness and efficiency, avoiding the conservatism of static robust control.

Each working condition category is mapped to a set of pre-set parameter values:

$$\Pi : C_k \rightarrow \left(\alpha_k, H_k, \Gamma_{Q,k}, \Delta I_{max,k}, R_k^{cell}\right), k = 0, 1, \ldots, 6 \tag{11}$$

Here, $\alpha_k$it is the error level of post-calibration,H_k is the rolling prediction-optimized time domain length, $\Gamma_{Q,k}$ is the uncertainty budget and $\Delta I_{max,k}$ is the upper limit of DC current ramping. The specific assignments for these parameters are determined through a hybrid strategy combining offline statistical analysis of historical data and expert process knowledge. The risk parameters for high-risk conditions (e.g., anode changing or instability) are calibrated based on the extreme disturbance boundaries observed in historical fault logs to ensure worst-case safety. Conversely, parameters for normal conditions are optimized via offline sensitivity simulations to maximize economic efficiency without violating soft constraints.

In summary, this section provides an integrated link from data, prediction, constraints, robust solution, and working condition triggering. To validate improvements in energy efficiency, carbon intensity and operational costs, this section conducts benchmarking and comparative experiments using industrial-grade data for verification. This method achieves these gains while satisfying strict constraints and ensuring feasibility.

3. Results and discussion

To demonstrate the feasibility of the proposed generalized framework, numerical validation is performed on a real-world 300 kA aluminum electrolysis plant in Yunnan Province, China. This facility represents a highly constrained scenario for industrial flexibility due to its high energy demand, strong electro-thermal coupling, and continuous operation boundaries. The empirical study utilizes 18 months of high-resolution operational data, comprising 5-min interval records of potline power, cell voltage, and auxiliary system loads. The system topology follows the instantiated EH structure described in Fig. 1. The specific process constraints defined in Section 2.3 are strictly enforced in the simulation to ensure that flexibility potential is evaluated under

Table 2
Classification of typical operating conditions, trigger criteria, and energy flow impact.

Event category	Specific condition	ID	Trigger criterion
Market Regime	Peak/Off-peak Period	C0, C1	High/Low Price
Internal Operation	Anode Changing	C2	Load Drop
Internal Operation	Pot Startup	C3	Current Ramping
Process Anomaly	Cell Instability	C4	Voltage Noise
Process Anomaly	PV Surge	C5	Power Mismatch
System Fault	System Fault	C6	Coupling Failure

realistic safety margins.

3.1. Quality of predicting intervals

To evaluate the quality of the prediction intervals, two standard metrics were employed under nominal coverage levels of 90% and 95%: the PICP and the MPIW. PICP represents the empirical proportion of observations contained within the constructed intervals, thereby assessing reliability with respect to the target confidence level. MPIW denotes the average width of the prediction intervals and serves as an indicator of sharpness; for a given coverage level, a smaller MPIW corresponds to a less conservative and more informative uncertainty representation. Monthly statistics for photovoltaic generation and heat load are reported with 95% confidence intervals.

Fig. 3 displays the monthly statistical performance for photovoltaic and thermal loads at nominal confidence levels of 90% and 95%. The results indicate that PICP aligns closely with the nominal targets. In particular, the coverage rates for photovoltaic generation reach 0.962 and 0.964 for the 90% and 95% intervals, respectively. Similarly, the thermal load maintains a coverage rate consistently above 0.997. Meanwhile, MPIW remains within a reasonable range, with values of approximately 5.3 MW for photovoltaic power and 1.0 MW for thermal load. These metrics confirm that the model provides compact and practical boundaries rather than achieving high coverage through excessively wide intervals.

Fig. 4 presents the constraint-violation frequency curves from 1 to 48 h. For PV, the 90% interval exhibits a default rate of approximately 2–6% and increases with step length during both the ramp-up and ramp-down phases. The 95% interval is uniformly lower and preserves the expected ordering between confidence levels. Thermal load is markedly more stable. Default rates remain below 1% at both confidence levels and variations across step sizes are modest. These observations agree with the monthly statistical results and support using the calibrated interval as a reliability constraint in subsequent MPC scheduling. Accordingly, the calibrated lower and upper bounds function as a probabilistic safety envelope. The 90% interval is adopted for routine daily operations to avoid excessive conservatism, while the 95% interval is applied during PV slope-trigger windows to mitigate short-term exceedance risk.

Given that industrial field data are costly to obtain, the method used here is naturally well suited to small samples. Conformal prediction requires no assumption about the error distribution and can produce prediction intervals with formal coverage guarantees even with limited data. Crucially, the inevitable trade-off between higher coverage and wider intervals is visible yet remains controllable. Under both nominal 90% and triggered 95% operating conditions, the empirical coverage

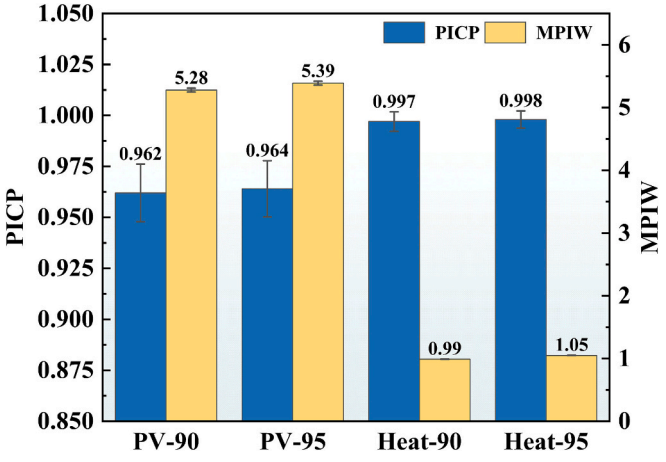


Fig. 3. Monthly statistical evaluation of prediction intervals.

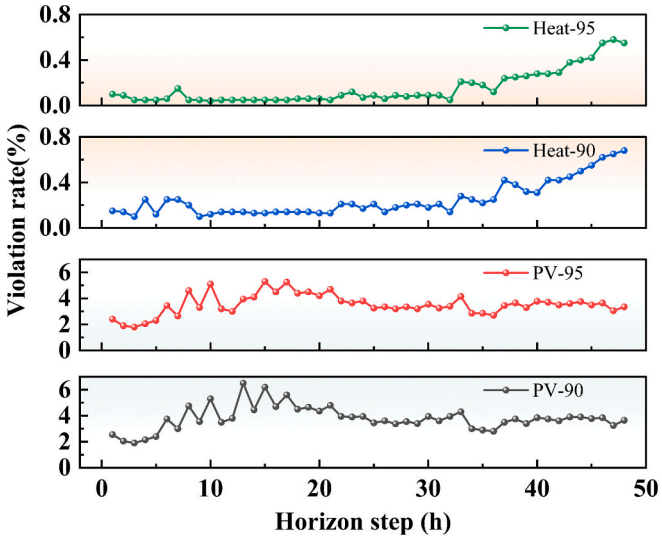


Fig. 4. Stepwise reliability diagnosis.

closely matches the target. This consistency provides an interpretable basis and reliable inputs for the subsequent robust MPC. This improvement in coverage stability and interval score aligns with findings in renewable-power forecasting, where conformal calibration attains nominal coverage without distributional assumptions [12].

3.2. Baseline multi-energy characteristics analysis

18 months of monthly records were used to establish a monthly baseline from the three dimensions of specific energy consumption (SEC), carbon intensity (CI) and unit cost (Cost), and achieve consistency in methodology without changing any historical power sequence. The calculation methodology is defined as follows. Energy intensity is derived from total electricity consumption and production output. Carbon intensity is calculated based on purchased electricity and monthly grid emission factors. Finally, unit cost consists of three components including the electricity energy fee, the maximum demand fee, and the carbon cost. The demand fee is billed based on the maximum demand, with a unit price of λ_{peak}=6.25 USD·kW^{-1}·month, calculated based on the monthly peak of the 15-min rolling average power. The carbon price and the monthly average electricity price are based on the median of the seasonal price band. Accordingly, the monthly operating cost identity is:

$$\text{OPEX}_m = \text{grid}_m \cdot \bar{p}_m + \lambda_{\text{peak}} \cdot P^{max}_{15\text{min},m} + \text{grid}_m \cdot EF_m \cdot \frac{p_{CO_2}}{1000},\ C^{\text{unit}}_m = \frac{\text{OPEX}_m}{m_{\text{Al},m}} \tag{12}$$

Fig. 5 traces the temporal evolution of unit electricity cost, CI, and SEC. Specifically, the solid lines represent the monthly average values, while the shaded bands indicate the standard deviation, reflecting the volatility of daily operations. Fig. 5 reveals both coupling and divergence among these metrics. Unit cost and CI display tightly synchronized U-shaped seasonal cycles. Both metrics fall to their minimum levels during the wet season from June to October. This decrease occurs because a higher share of clean energy in the Yunnan power grid lowers the grid emission factor. In contrast, both rise to peaks during the dry season from December to April of the following year. This strong waveform concordance implies that external hydrological cycles, rather than internal process variability, primarily drive economic and environmental cost changes. By contrast, SEC shows only minor fluctuations throughout the observation period and remains confined to a narrow band around 13.2 ± 0.2 MWh·t^{-1}Al, with no clear seasonal pattern. This stability indicates that the bulk energy efficiency of the electrolyzer has

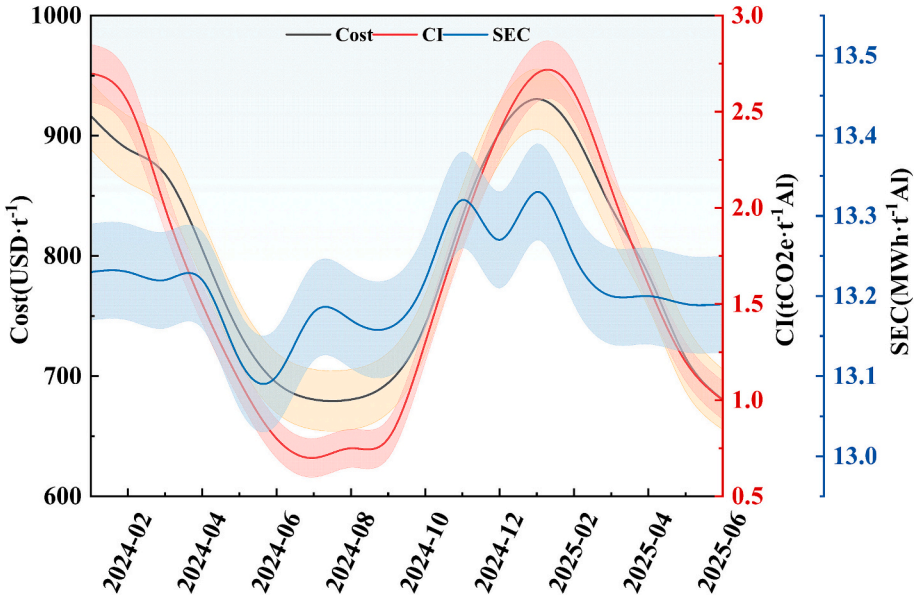


Fig. 5. Baseline monthly intensity and cost metrics.

reached a physically steady state under current process control. Fig. 5 presents a multidimensional comparison. It reveals that SEC remains relatively rigid. As a result, shifting load across multi-energy systems is the most effective strategy. This approach targets periods of simultaneously high costs and high CI. By doing so, it maximizes both cost savings and emission reductions.

Fig. 6 presents row-normalized heat maps. These maps illustrate the temporal dynamics of energy charges, demand charges, and carbon costs across hydrological cycles. Normalization removes scale differences between these metrics. This normalization highlights the sensitivity of each component to seasonal variation. Although energy and electricity charges dominate in absolute terms, their normalized intensity varies only modestly over the cycle. By contrast, carbon costs undergo the largest seasonal swings. In the wet season, carbon cost intensity falls into the blue trough. In the dry season, rising grid emission factors propel it into the orange peak. Demand charges and carbon costs display pronounced high intensity coupling in the dry season. As the bright-yellow regions indicate, periods of elevated emission factors frequently coincide with peak demand-charge pressure. Thus, even though carbon costs represent a limited share of total costs in absolute terms, their temporal overlap with demand charges magnify the risk of higher unit operating costs during dry periods.

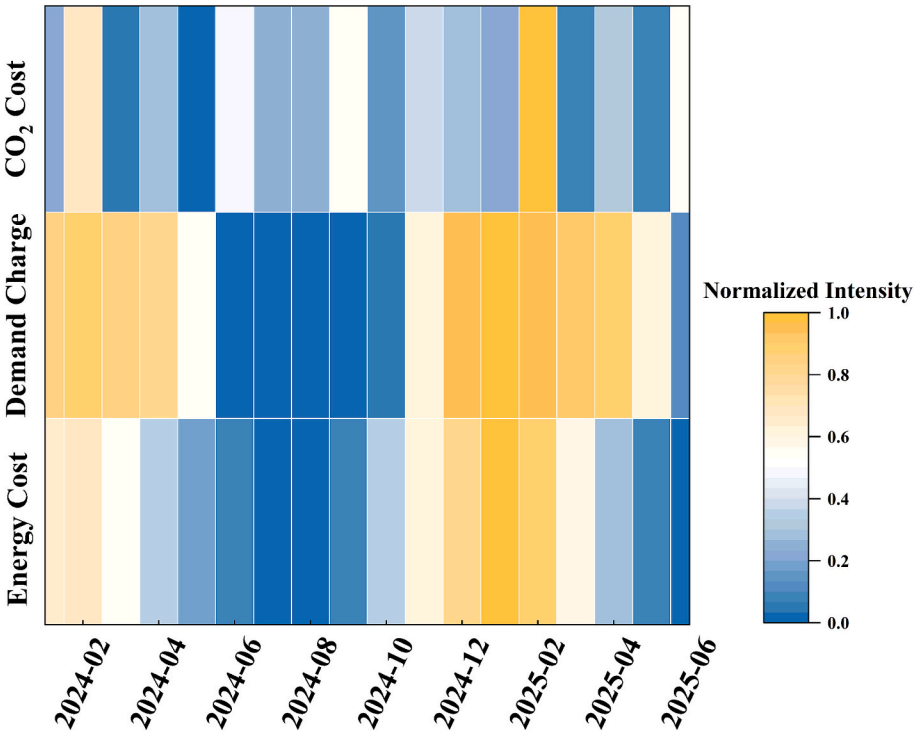


Fig. 6. Seasonal characteristics of baseline energy efficiency and carbon cost metrics.

3.3. Control performance under varying conditions

3.3.1. Dynamic control performance under time-varying conditions

Fig. 7 shows the temporal variation of the single-day grid emission factor EF_{grid}. The simulation maintains strict plant-level operating constraints. Three control strategies are compared: D-MPC ($\Gamma \approx 0$), S-RMPC (fixed), and CT-RMPC (adaptive to events). The evaluation metrics cover the power purchase trajectory, peak shaving capability, and default frequency. They also include unit product carbon cost and carbon intensity.

Fig. 8 displays the results for D-MPC, which ignores EF(t) and therefore continues to consume high power during periods of elevated emission factors. S-RMPC reduces peak loads by maintaining a fixed conservatism level, at the expense of inefficiency during non-event intervals. In contrast, CT-RMPC adaptively adjusts the robustness budget upon detecting operating condition transitions (e.g., C2 events or abnormal residual surges). Through coordinated HP–TES charge–discharge scheduling, power purchases are strategically shifted away from high-EF periods while avoiding unnecessary conservatism in stable intervals.

Beyond economic performance and operational feasibility, the trajectory-level characteristics in Fig. 8 provide quantitative insight into grid-service flexibility. The visibly moderated slopes reflect explicitly constrained ramping imposed by electro-thermal stability limits of the electrolytic cells, ensuring that regulation remains within the allowable operational envelope. The maximum observed ramping magnitude under CT-RMPC remains within approximately 3 MW per 5-min interval, which is substantially lower than typical industrial rectification ramping capabilities and therefore consistent with safe electro-thermal operation. Rather than frequent discrete switching, flexibility is realized via gradual trajectory shaping enabled by auxiliary subsystem coordination. The sustained evening peak suppression indicates the controller's capability to maintain regulation over extended time horizons, consistent with medium-timescale reserve provision. Combined with the decreasing violation frequency observed under increasing robustness budgets, these results demonstrate that achievable flexibility is characterized by stable, bounded, and prolonged modulation, aligning with peak shaving and reserve-type services while preserving operational integrity.

Sensitivity analysis shows that the peak grid power purchase falls monotonically as the uncertainty budget Γ increases, with the CT-RMPC method producing a steeper decline. Raising Γ from 0 to 2 reduces the daily peak load from 160 MW to 151 MW and lowers the daily constraint-violation frequency from 4 to 0.2. Beyond $\Gamma \approx 2$, however, the marginal peak-shaving benefit diminishes rapidly. Meanwhile, the TES *SoC* trajectory develops pronounced high-frequency oscillations, indicating that the controller becomes overly conservative. The analysis balances peak suppression and the stability of the *SoC*. Thus, it identifies $\Gamma \approx 2$ as the optimal parameter. This value represents the optimal balance between robustness and economic performance. To alleviate concerns regarding equipment lifetime under frequent regulation, it is noted that power modulation is strictly bounded by the ramping constraints of the electrolytic cells, which prevent abrupt electro-thermal disturbances. Moreover, the Γ-scan analysis indicates that overly conservative settings may induce high-frequency actuation without additional economic benefits, thereby motivating the selection of $\Gamma \approx 2$ as a compromise between robustness and smooth operation. Consequently, the achieved flexibility is characterized by gradual trajectory shaping rather than frequent discrete switching, which supports long-term operational integrity.

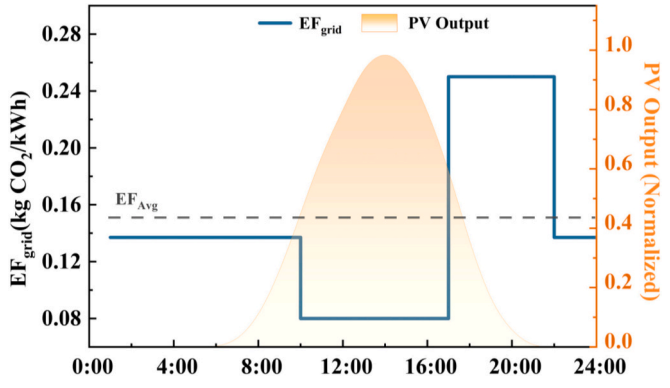


Fig. 7. Hourly dynamics of EF_{grid} superimposed with PV generation profile.

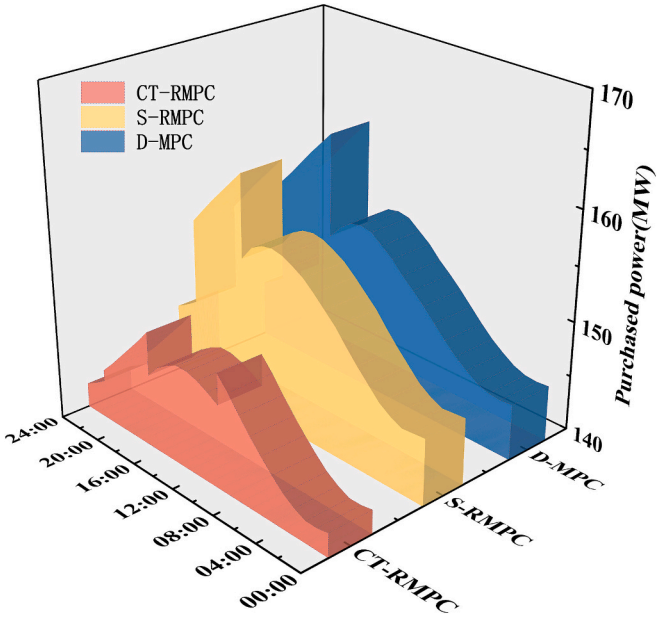


Fig. 8. Purchased power trajectories of three control strategies.

In terms of environmental performance, the CT-RMPC strategy reduces the indirect carbon emission intensity of molten aluminum by reoptimizing intraday multi-energy scheduling. Unlike D-MPC, which ignores time-varying emission factors, and S-RMPC, which applies a uniformly conservative robustness policy, CT-RMPC exploits the intraday variability of the grid emission factor, EF_{grid}, and concentrates the robustness budget on periods with high EF. By doing so, it curbs grid electricity purchases during high-carbon intervals while preserving a steady molten-aluminum output, and it shifts energy replenishment to low-carbon periods through coordinated operation of the HP and TES. Consequently, CT-RMPC effectively decouples the production load from high-emission windows and achieves lower product-level carbon intensity than both benchmark strategies, indicating improved compliance readiness under increasingly stringent carbon constraints.

3.3.2. Mechanism of adaptive robustness and parameter optimization

Based on the interval calibration detailed in Section 3.1, this analysis maps the conformal intervals onto the robust MPC framework to examine scenarios with significant operating fluctuations. A representative 160 MW electrolyzer group was selected as a case study, where the benefits and costs of the triggering mechanism were evaluated through Γ scans and monthly-scale statistics.

Fig. 9 shows that as Γ increases from 0, both the plant-level peak value and operating cost drop significantly before stabilizing. The Γ scan indicates a first inflection at $\Gamma \approx 2$. At this specific value, the plant-level peak declines from 159.6 MW to 157.9 MW, a peak shaving of about 1.7 MW. Further increases in Γ yield progressively smaller benefits, reflecting diminishing marginal returns. In this regime, flexible resources cycle more frequently, but the *SoC* does not reach its upper or lower bounds. The framework handles uncertainty and reduces peaks by

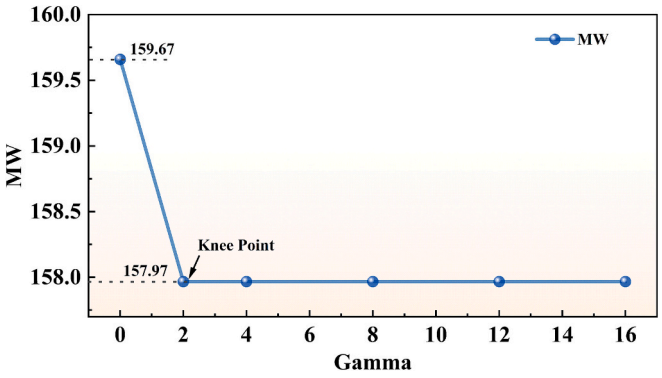


Fig. 9. Γ -scan on systemic grid peak.

coordinating the HP and TES, all while keeping the electrolyzer load constant.

To reveal the coupling mechanism between the external market environment and internal control strategies, this work selects two key variables, namely the demand billing coefficient λ_d and the uncertainty budget Γ. λ_d is a dominant factor in system returns. Accordingly, Fig. 10 presents the constructed response surface of the NE, which illustrates the nonlinear impact of these dual incentive and control factors on system income.

Along the λ_d-axis direction, NE shows a significant linear growth trend. These results indicate that in an industrial scenario with a large installed load capacity such as electrolytic aluminum, the marginal benefits brought by the HP-TES system in reducing maximum demand through peak shaving and valley filling far exceed simple energy time-shift arbitrage. A higher λ_d means that the power grid has stronger incentives for load stabilization, which significantly amplifies the economic leverage effect of this strategy.

Along the Γ-axis, NE rises sharply as Γ increases from 0 to 2, indicating that introducing a moderate robustness budget effectively mitigates default penalties from prediction errors. Beyond $\Gamma \approx 2$, the surface flattens and saturates, suggesting that further increases yield diminishing returns. This knee point identifies $\Gamma \approx 2$ as the optimal trade-off between economic performance and operational safety, as it provides a practical compromise between economic performance and operational safety for the studied plant, improving robustness to data-observed disturbance realizations while avoiding the conservative losses associated with excessive budgets. This behavior aligns closely with findings in comprehensive energy-system flexibility studies. For instance, in

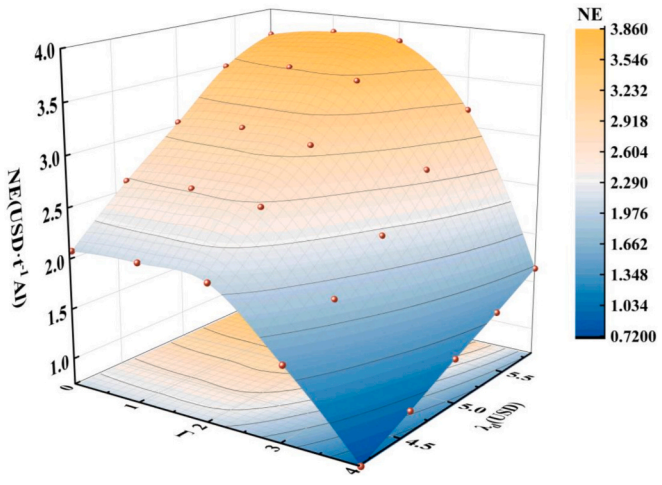


Fig. 10. Response surface of annualized NE regarding λ_d and uncertainty budget Γ.

multi-energy systems with thermal inertia, Li et al. demonstrated that the relationship between the uncertainty budget and operating cost is a nonlinear Pareto trade-off, where a moderate uncertainty budget captures most renewable fluctuation risks, but an excessive budget wastes flexibility and degrades overall economics [46]. The surface curvature shows that in the low λ_d region, improvements in Γ contribute little to NE, whereas in the high λ_d region the unit budget benefit of Γ increases substantially. Thus, precise and robust control is essential to ensure profitability during dry periods or when electricity prices and demand cost are high.

In summary, the response surface offers clear practical guidelines for implementation. The economic advantage of the strategy peaks when demand fees are high. To balance robust operation with economic performance, the results suggest a configuration of $\Gamma \approx 2$.

3.3.3. Statistical validation of long-term resilience

While the single day analysis demonstrates the superior dynamic performance of the proposed strategy, evaluating its statistical consistency over a longer horizon is essential to rule out randomness driven by specific weather or market conditions. As shown in Fig. 11, a pairwise distribution comparison of monthly operations is conducted to verify the long-term robustness of the strategy.

First, regarding grid interaction, Fig. 11(a) illustrates the distribution of peak shaving capability. The results indicate that the CT-RMPC effectively mobilizes the latent flexibility of the auxiliary system. In the trigger mode, the median peak shaving significantly increases to 1.23 MW relative to nominal control. This improvement demonstrates that the adjustment depth approximately doubles the baseline capability, corresponding to 0.8% of the series nominal load. This quantitative evidence confirms that the condition-triggered mechanism effectively mitigates the excessive conservatism of traditional robust control, enabling the system to provide significant regulation capacity during critical peak windows.

Secondly, regarding economic viability, Fig. 11(b) confirms that the operational advantages translate into consistent financial returns. The monthly cost savings consistently range between 40,300 USD and 59,700 USD. Furthermore, rigorous statistical tests, including the Wilcoxon signed-rank test and effect-size analysis, confirm that this profitability is statistically significant across different statistical thresholds and is not an artifact of data outliers. This consistency suggests that the strategy can achieve robust benefits even under highly volatile market environments.

Finally, Fig. 11(c) reveals the internal operational logic underpinning these external gains. The trigger mode exhibited higher variability with a range of approximately 0.04. This significantly exceeded that of the nominal mode. However, this increased variability did not compromise safety. The system state remained strictly within operational constraints throughout. This observation validates the core design principle. The controller proactively leverages the thermal inertia of the TES to absorb disturbances. Finally, the system successfully secures overall economic benefits by trading static safety margins for dynamic regulation capabilities.

3.4. Long-term economic robustness and parameter sensitivity

3.4.1. Seasonal adaptability and positive net effect

To evaluate long-term economic robustness, the cross-month NE is computed by applying the trigger mechanism to the full 18-month dataset. Since monthly variations in liquid aluminum production shift the base load and render fixed thresholds prone to false positives or missed triggers, a dimensionless scaling approach based on the load rate is introduced. In particular, the algorithm employs the monthly quantile band as a benchmark curve, dynamically scaling it according to the monthly average power ratio.

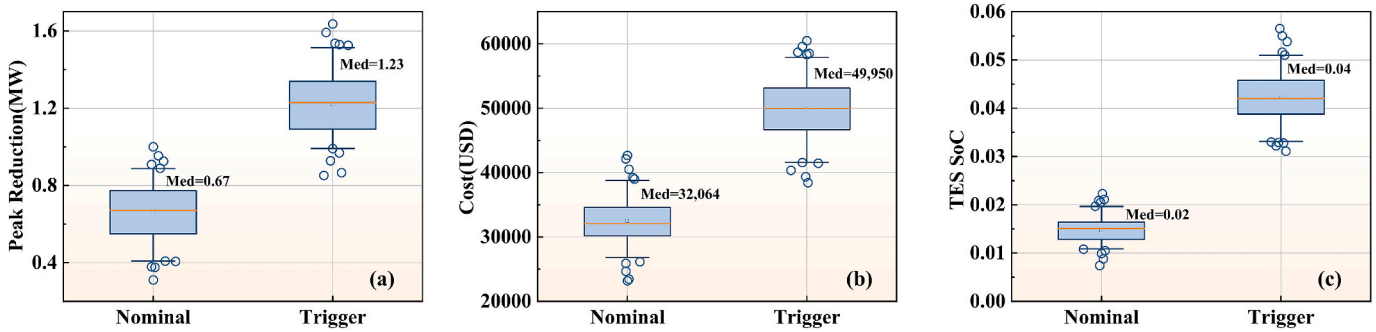


Fig. 11. Pairwise distribution comparison of monthly peak shaving capability (a), cost reduction (b), and TES state variability (c) between Nominal and Trigger control modes.

$$\Delta P95_m^{(q)} = \Delta P95_{\mathrm{ref}}^{(q)}\left(I_m / I_{\mathrm{ref}}\right)$$
$$\Delta E_m^{(q)} = \Delta E_{\mathrm{ref}}^{(q)}\left(\frac{E_m}{E_{\mathrm{ref}}}\right) \tag{13}$$

where I_m is approximately 0.9 as the steady-state benchmark, $P_m^{\max}$ is the monthly 15-min peak value, and $\overline{P}_m$ is the monthly average power. This ensures that the control strategy floats adaptively with the production base. Accordingly, the monthly NE comprises three components. These are energy charge savings, demand charge reductions, and carbon cost savings:

$$NE_m^{(q)} = \Delta E_m^{(q)}\overline{p}_m + \kappa\Delta P95_m^{(q)}\lambda_{\mathrm{peak}} + \Delta E_m^{(q)}EF_m\frac{p_{CO_2}}{1000} \tag{14}$$

Fig. 12 shows the evolution trajectory of cross-month NE with the hydrological cycle of Yunnan Province from January 2024 to June 2025. The data points in Fig. 12 illustrate the central trend of monthly returns, while the error bars depict the dispersion arising from production fluctuations and operational variability. The NE curve exhibits seasonal patterns driven by the hydrological cycle of Yunnan Province. During the dry season, higher electricity prices and larger emission factors increase the marginal value of system flexibility. Thereby enabling deep peak shaving via HP-TES to maintain the NE in the 2–4 USD·t^{-1}Al range. In the wet season, higher hydropower output reduces the peak–valley price spread and lowers the emission factor, which compresses energy-arbitrage opportunities. Nonetheless, by adapting based on Eq. (14), the strategy captures modest demand-side benefits and maintains the NE consistently above 0.4 USD·t^{-1}Al.

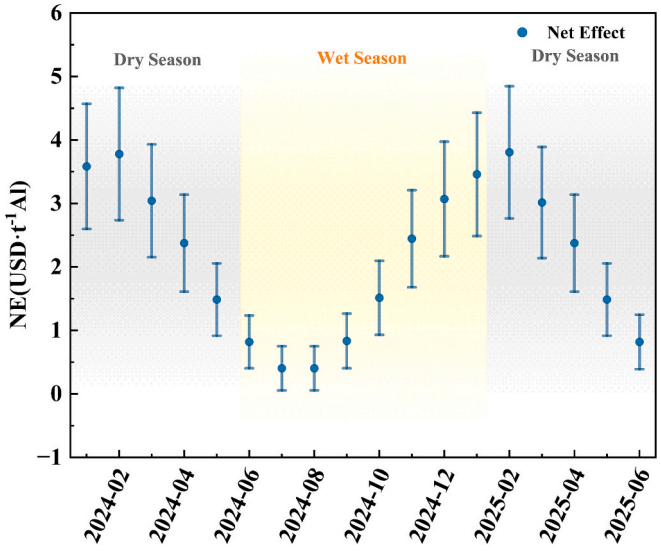


Fig. 12. Monthly variations of the net effect.

Incorporating actual production data introduces fluctuations into the benchmark. However, the lower bounds of the error bars remained strictly positive across all months. This confirms that the strategy consistently yields positive returns. It performs well during both the wet season with volatile production and the dry season with high costs. By dynamically adjusting the uncertainty budget, the system effectively eliminates the risk of operational loss.

3.4.2. *Sensitivity to internal control parameters*

Beyond external seasonal adaptability, the engineering viability of the strategy also relies on its insensitivity to internal parameter tuning. To evaluate the stability of the CT-RMPC framework under control-parameter disturbances, multidimensional sensitivity experiments are conducted targeting the trigger threshold, energy replenishment margin, and uncertainty budget. Utilizing 18 consecutive months of measured data, full-cycle simulations are executed repeatedly across various parameter combinations to assess performance variations.

First, the central trend of NE proves robust across all tested parameter combinations. Under all operating conditions, the monthly median NE displays a U-shaped profile that inversely mirrors the hydrological cycle in Yunnan Province, while remaining positive across the entire parameter space. This consistency confirms that economic gains stem from exploiting peak–valley price spreads and the demand tariff mechanism rather than overfitting to a specific parameter set. Adjusting parameters merely shift the curve within a limited range without altering its overall form.

Secondly, the stringency of parameter thresholds mainly affects the dispersion of returns. For instance, raising the trigger condition from P95 to P97 effectively filters out false triggers caused by random disturbances. Reducing the replenishment limit from 2.0% to 1.5% yields a similar effect. These adjustments significantly narrow the confidence interval of monthly net earnings. They also raise the lower profit limit under extreme operating conditions. Furthermore, the choice of Γ reflects the trade-off between risk and return. A lower Γ may marginally improve theoretical earnings. However, it increases the probability of constraint violations and *SoC* oscillation. In contrast, a higher Γ prioritizes operational security by accepting minor economic concessions.

In summary, the internal parameters primarily act as fine adjustments that regulate risk exposure and return volatility without undermining the fundamental economic viability of the strategy. Across a reasonable engineering range, the strategy preserves positive effects and a stable seasonal pattern under all tested parameter combinations, indicating strong robustness of engineering. Consequently, practical deployment can achieve reproducible gains without necessitating labor-intensive point-by-point calibration.

3.4.3. *Robustness analysis and method applicability*

Effect size analysis is employed to reveal the underlying mechanism that enables the CT-RMPC strategy to maintain stable returns despite output fluctuations. The analysis quantifies the contribution of different

return components to the NE. Fig. 13 presents an effect-size analysis using Cliff's δ for energy charges and demand charges before and after implementing the strategy. To facilitate interpretation, dashed horizontal lines delineate standard thresholds where 0.147 indicates small effects and 0.33 marks the boundary for medium effects. The effect size for demand charges is significantly larger than the other two cost components and consistently exceeds the medium threshold, indicating a dominant impact. Meanwhile, the effect size for energy charges is smaller but still noticeable, generally hovering around the small-effect threshold. Carbon cost changes are the smallest, although they follow the same directional trend as the other metrics. Consistent with the operational analysis in Section 3.3, these results demonstrate that the strategy derives its primary economic advantage from peak shaving. This effectively mitigates demand charges. In parallel, secondary benefits arise from load shifting behaviors that exploit time-of-use prices and carbon cost signals.

This attribution analysis also explains the insensitivity of the strategy to internal parameters. Since the dominant return stems from capturing critical peak moments rather than from precise adjustments to total energy consumption, the system can achieve robust profitability without requiring overly sensitive parameter tuning. This characteristic confirms the strong engineering robustness of the proposed framework. Thus, the system is well-suited for industrial scenarios where operational reliability during peak suppression is a priority. Compared to precise yet high-risk energy arbitrage, such stability is prioritized.

3.5. Decarbonization potential under dynamic pricing

The preceding benefit evaluations are primarily predicated on the current monthly settlement mechanism. However, as electricity markets evolve towards higher temporal resolution, capturing intraday real-time carbon price signals has emerged as a critical pivot for the deep decarbonization of EIIs [47]. To validate the adaptability of the proposed framework within prospective market environments, this section incorporates hourly-varying grid emission factors, EF(t), alongside real-time carbon pricing signals. The objective is to demonstrate the capability of the framework to autonomously adjust multi-energy flow strategies in response to dynamic carbon signals, thereby achieving an optimal trade-off between low-carbon objectives and economic viability.

Based on the typical operational characteristics of the Yunnan power grid during a representative month, the hourly grid emission factor curve is illustrated in Fig. 7. By coupling this factor with real-time grid

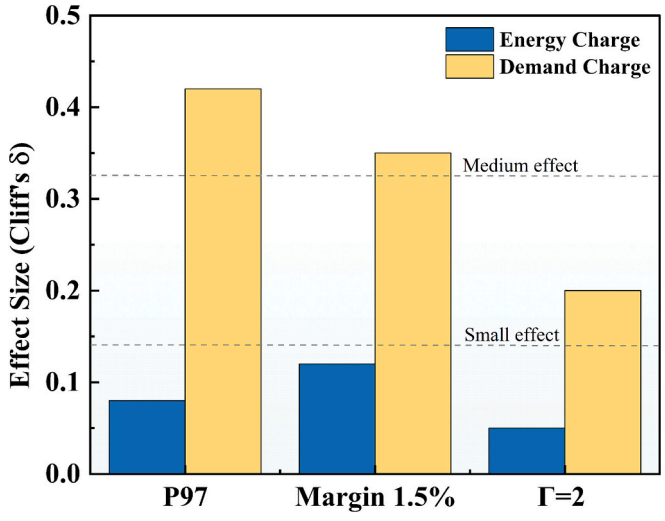


Fig. 13. Effect size comparison between energy and demand charges across different sensitivity scenarios.

power consumption p_{CO_2}, a carbon cost term, $Cost_{CO2}(t)$, is formulated and incorporated into the MPC objective function. The simulation results in Fig. 14 elucidate the adaptive decision-making mechanism of the CT-RMPC strategy under multiple constraints. Upon the introduction of real-time carbon signals, the controller exhibits distinct synergistic behavior among multiple objectives.

After the introduction of real-time carbon cost signals, the behavior of the MPC controller changes significantly, showing a strong tendency for low-carbon optimization. In contrast to the baseline thermal-load-following strategy, the MPC actively ramps up HP power to rated capacity (>8.0 MW) during low-carbon windows, thereby executing carbon-aware load shifting across intraday fluctuations. This represents a strategic trade-off involving thermal energy storage conversion losses. Table 3 quantifies this process where the round-trip efficiency loss incurs an increase in Energy Charges of approximately 22,200 USD. However, this operational cost is effectively leveraged to achieve a far greater reduction in Demand Charges amounting to 73,800 USD. Although the current carbon revenue of approximately 1100 USD is only a marginal contribution, the system achieves a monthly net surplus of 52,700 USD. This financial analysis demonstrates that the core profit of the strategy is to exchange low energy loss for high-capacity resource value.

Currently, real-time carbon pricing is not yet widely implemented in domestic or regional power grids. Therefore, the preceding analysis should be viewed as a scenario-based assessment. The simulation results highlight two primary conclusions. First, incorporating high-resolution emission factors directly into control targets empower electrolytic aluminum plants to manage electricity costs and carbon prices within a unified optimization framework. This integration naturally facilitates the coordination of peak–valley load shifting with emission reduction goals. Second, within a reasonable range of carbon prices, emission reductions do not necessarily conflict with economic performance. Instead, an optimal equilibrium emerges to balance operating costs and carbon intensity. Collectively, these results provide a robust technical framework. They support the development of future scheduling strategies and incentive mechanisms for electrolytic aluminum multi-energy systems that leverage real-time carbon signals.

3.6. Engineering implementation and policy implications

Based on the multidimensional analysis above, this section outlines an implementation roadmap for the CT-RMPC framework from engineering deployment and institutional design perspectives. At the engineering level, the study proposes a deployment paradigm centered on streamlined parameterization. In practice, this entails adopting a robust baseline configuration that suppresses demand peaks while mitigating frequent hardware cycling. Comparative studies show that, for identical budgets, event-triggered robust strategies yield lower peak violation rates and more stable cost control than traditional static robust methods [48,49]. Building on this finding, the scheduling logic should concentrate on high-value time windows and prioritize HP-TES for peak shaving and valley filling. Engineering simulations confirm that the coordinated strategy of pre-peak charging and peak-period discharging improves both cost and CO_2 reduction under time-varying price and emission factor constraints [48]. The framework is also highly transferable in a non-intrusive manner. Coupling heat pumps with thermal energy storage enables effective load shifting and demand suppression without altering the main process boundaries, such as the thermal balance of electrolytic cells [50]. Therefore, deployment can often be achieved via local data re-parameterization and supervisory integration with existing automation; the required level of controller refactoring depends on plant-specific instrumentation, safety interlocks, and legacy control architectures.

Beyond technical measures, policy frameworks need to foster an institutional environment with aligned incentives to enable flexible deployment. It is recommended to maintain a price signal framework

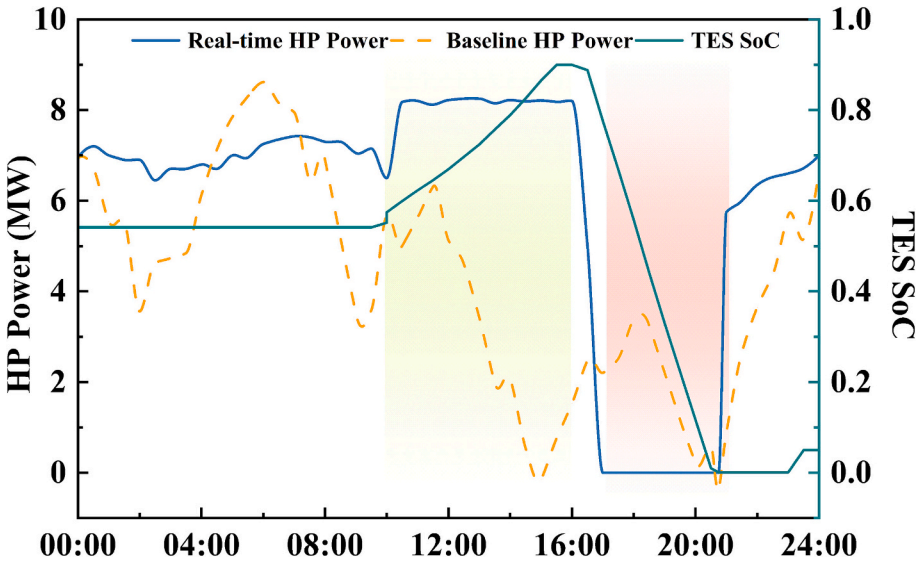


Fig. 14. Comparison of operational strategies for heat pumps and thermal storage systems driven by real-time carbon price signals.

Table 3
Breakdown of monthly financial trade-offs under real-time carbon price signals.

Component	Mechanism	Financial impact (USD)	Contribution status
Carbon-cost saving	Intraday low-carbon arbitrage	+1100	Marginal Gain
Energy Charge	Round-trip efficiency loss (TES)	−22,200	Cost Penalty
Demand Charge	Coordinated peak shaving	+73,800	Dominant Gain
Net Effect	Comprehensive Surplus	+52,700	Positive Profit

anchored by demand charges, while introducing a flexible response waiver mechanism to mitigate corporate concerns regarding instantaneous load rebounds. At the same time, the pilot release of hourly grid emission factors is initiated during a critical period. Studies indicate that within an hourly-EF framework, the higher carbon intensity caused by reservoir dispatch in dry seasons can be partly offset by temporal load shifting, thereby aligning economic objectives with low carbon targets and encouraging firms to consume cleaner energy [51]. Finally, implementing comprehensive data logging establishes a verifiable evidence chain. This provides a robust audit basis for future carbon market compliance and ancillary service compensation.

Although demonstrated on aluminum electrolysis, the proposed condition-triggered robust MPC framework is potentially transferable to other EIIs with regime-dependent feasibility envelopes and inertia-coupled dynamics; however, transfer requires process-specific model instantiation, constraint re-identification, and site-level validation. Many industrial assets provide natural flexibility windows through staged operation and buffering dynamics, enabling short-term load reshaping without violating safety envelopes. For example, steelmaking units often feature distinct high-power and low-power operating stages, while rotary kilns in cement and glass manufacturing possess substantial thermal inertia that can temporally buffer energy inputs. These characteristics are analogous to the operating-regime switching and inertia-driven buffering exploited in the proposed condition-triggered robust MPC. Accordingly, the developed framework offers a generalizable decision-support paradigm for coordinating industrial flexibility provision with feasibility-guaranteed operation under uncertainty.

4. Conclusion

To reconcile the inherent rigidity of EIIs with the increasing flexibility demands of high-renewable grids, this work proposes a generalized CT-RMPC framework. By synergizing an electro-thermally decoupled EH model with split conformal prediction, the strategy employs a hierarchical trigger mechanism to dynamically update the uncertainty budget, thereby optimizing the critical trade-off between process safety and economic efficiency in real-time. Empirical validation using 18 months of high-resolution industrial data from a 300kA aluminum smelter demonstrates that the strategy successfully unlocks the latent flexibility of auxiliary systems, reducing the daily peak grid load from 160 MW to 151 MW. While providing this grid support, the process state variables are maintained within the safety envelope, with the SEC stabilized at 13.2 ± 0.2 MWh·t^{-1} Al, demonstrating that flexibility is not achieved at the expense of production stability. Furthermore, the framework achieves a cumulative cost saving of 652,000 USD over the validation period, with demand charge reduction contributing over 35% to the net effects. Sensitivity analysis identifies $\Gamma \approx 2$ as the optimal knee point, where constraint-violation frequency drops to nearly zero without inducing excessive hardware cycling. Ultimately, this study proves that by integrating process-aware safety mechanisms with data-driven uncertainty quantification, even the most safety-critical industrial loads can serve as reliable flexibility resources for the low-carbon energy transition. This paradigm shift paves the way for EIIs to evolve from passive consumers into active, stabilizing nodes in future smart energy systems.

CRediT authorship contribution statement

Jinbo Nie: Writing – original draft, Methodology, Investigation, Conceptualization. **Qiaojie Bai:** Methodology, Investigation. **Congwei Tang:** Methodology, Investigation. **Jinlong Du:** Writing – review & editing. **Jianhang Hu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Wei Cheng:** Data curation. **Cheng Tan:** Writing – review & editing. **Ruijin Fan:** Writing – review & editing. **Yong Yu:** Writing – review & editing. **Wanzhang Yang:** Writing – review & editing. **Hua Wang:** Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw operational dataset of the industrial aluminum electrolysis plant contains commercially sensitive information and cannot be made publicly available due to confidentiality agreements with the data owner. De-identified and aggregated data that support the main findings of this study (including the monthly energy–carbon–cost indicators and the data used to generate the figures) as well as the optimization scripts are available from the corresponding author upon reasonable request and with permission from [Yunnan Aluminum Co., LTD, Kunming 650,502, PR China].

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